Double Integrals

Section 1: Iterated Integrals over Rectangles

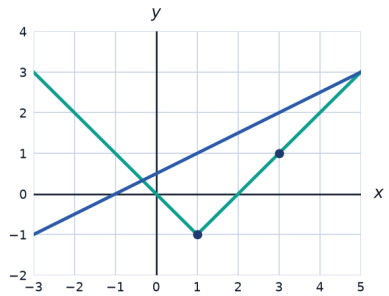
1 Evaluate $\int_0^2 \int_0^3 (xy^2)\,dx\,dy$.

a. 12

b. 18

c. 9

d. 6



2 Evaluate $\int_0^1 \int_0^2 (2x + 3y)\,dy\,dx$.

a. 8

b. 6

c. 10

d. 4

3 What does Fubini's Theorem guarantee for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. $\int_a^b \int_c^d f\,dy\,dx = \int_c^d \int_a^b f\,dx\,dy$

b. The integral equals the perimeter times height

c. The order of integration cannot be swapped

d. The integral must be zero

Section 2: Double Integrals over General Regions

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4 Evaluate $\iint_D (x+y)\,dA$ where D is bounded by $y=0$, $y=x$, and $x=1$.

a. $1/2$

b. $1/3$

c. $2/3$

d. $1/6$

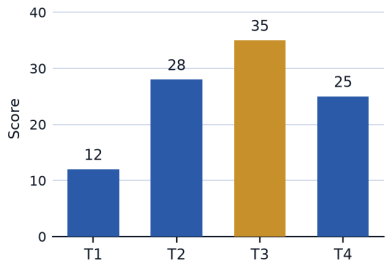
5 Reverse the order of integration for $\int_0^1 \int_x^1 f(x,y)\,dy\,dx$.

a. $\int_0^1 \int_0^y f(x,y)\,dx\,dy$

b. $\int_0^1 \int_y^1 f(x,y)\,dx\,dy$

c. $\int_0^1 \int_0^1 f(x,y)\,dx\,dy$

d. $\int_x^1 \int_0^1 f(x,y)\,dx\,dy$



6 Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. e^2-1

Section 3: Double Integrals in Polar Coordinates

7 What is the area element dA in polar coordinates?

a. $dA = r\,dr\,d\theta$

b. $dA = dr\,d\theta$

c. $dA = r^2\,dr\,d\theta$

d. $dA = \frac{1}{r}\,dr\,d\theta$

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8 Evaluate $\iint_R (x^2 + y^2)\,dA$ over the disk $x^2 + y^2 \le 4$.

a. 8π

b. 16π

c. 4π

d. 32π

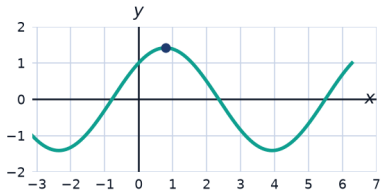
9 Evaluate the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx$ using polar coordinates.

a. $\sqrt{\pi}$

b. π

c. $2\sqrt{\pi}$

d. $\pi/2$



Section 4: Volume, Mass, and Center of Mass

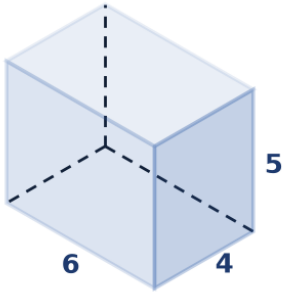
10 Find the volume under the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane.

a. 8π

b. 16π

c. 4π

d. 2π



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11 A lamina occupies the region $[0,1] \times [0,2]$ with density $\rho(x,y) = 3x$. Find its total mass M.

a. 3

b. 6

c. 1.5

d. 4.5

12 State the formula for the center of mass $\bar{x}$ of a lamina with density $\rho(x,y)$ and total mass M.

a. $\bar{x} = \frac{1}{M}\iint_D x\rho(x,y)\,dA$

b. $\bar{x} = \frac{1}{M}\iint_D y\rho(x,y)\,dA$

c. $\bar{x} = \iint_D x\,dA$

d. $\bar{x} = \frac{M}{\iint x\,dA}$

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